

PROGRESS REPORT · 2026

Adaptive Delivery of LLM-Generated Diagnostic Test Recommendations

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Reproducing & extending Nori et al. (2025) · SDBench + MAI-DxO

THE PROBLEM

The constraint is delivery, not diagnosis

- LLMs already match or exceed physicians on diagnostic reasoning
- **Goh 2024 (RCT):** the model alone beat physicians – but physicians *given* the model did not improve
- Indiscriminate alerts → alert fatigue → overrides. A great recommendation, poorly delivered, captures little value

THE QUESTION

Does more orchestration actually help?

Adaptive Alert Delivery · Recommendation systems

Three designs, one ladder of orchestration

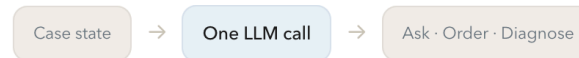
The same five clinical roles throughout – only how they are invoked changes.

BASIC

● Single LLM

1 model call / turn

Nori et al. generalist baseline.

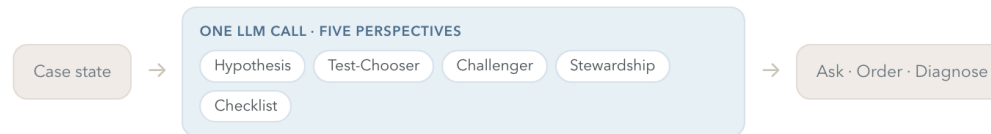


MEDIUM

● Roles prompt

1 model call / turn

The paper's MAI-DxO orchestrator: one call role-plays all five personas.

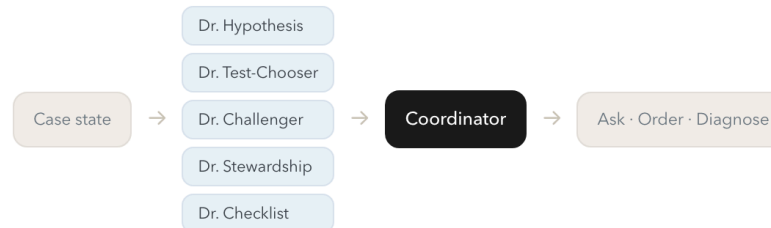


COMPLEX

● Multi-agent

≈ 6 model calls / turn

Our extension beyond the paper: each persona is its own agent, reconciled by a coordinator.



Same five MAI-DxO roles throughout – only the orchestration changes

Each case is worked step by step through a gatekeeper

Adaptive Alert Delivery · Recommend

MODEL llama3.1:8b · Single LLM

Case NEJMcp030015 · CPC · presentation only

View full case

A 38-year-old woman was admitted to the hospital in early August 2002 because of fever and confusion.

GK

CSF – protein 120 mg/dL, glucose 45 mg/dL, 200 cells/μL (80% lymphocytes). Opening pressure normal.

Order

Lumbar puncture

AI

GK

Normal CT scan of the brain.

Order

CT scan of the head

AI

GK

Positive for IgM antibodies against West Nile virus.

Order

Arbovirus panel

AI

GK

West Nile virus PCR on CSF: Positive.

Order

West Nile virus PCR on CSF

AI

J

Judge 4 / 5 · counts as correct

Core disease correct; a secondary qualifier is missing (meningoencephalitis implies inflammation of both meninges and brain), but management is unchanged.

Diagnosis

West Nile virus encephalitis

AI

Ground truth, withheld during the encounter: West Nile virus meningoencephalitis · 8 turns · 4 tests · \$1,210

The agent sees only the presentation; the Judge grades against the withheld truth

KEY RESULT

More orchestration does not buy accuracy

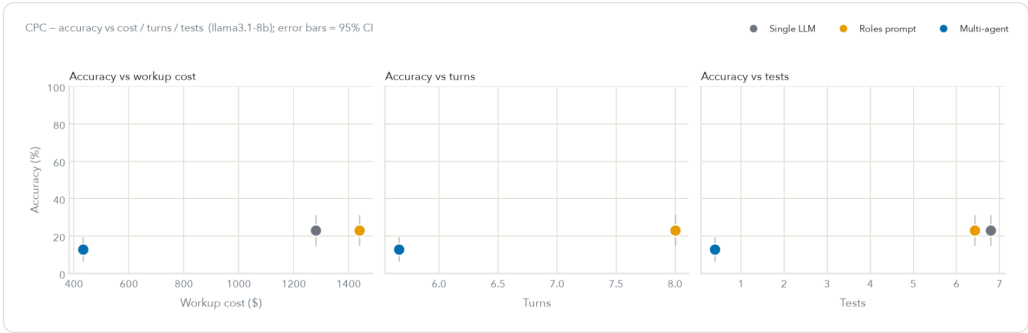
Adaptive Alert Delivery - Results

MODEL llama3.1:8b

CPC benchmark – accuracy vs. cost of orchestration

Llama-3.1-8B · n = 100 · same five clinical roles throughout · error bars are 95% CI

- Single LLM · basic
- Roles prompt · medium
- Multi-agent · complex



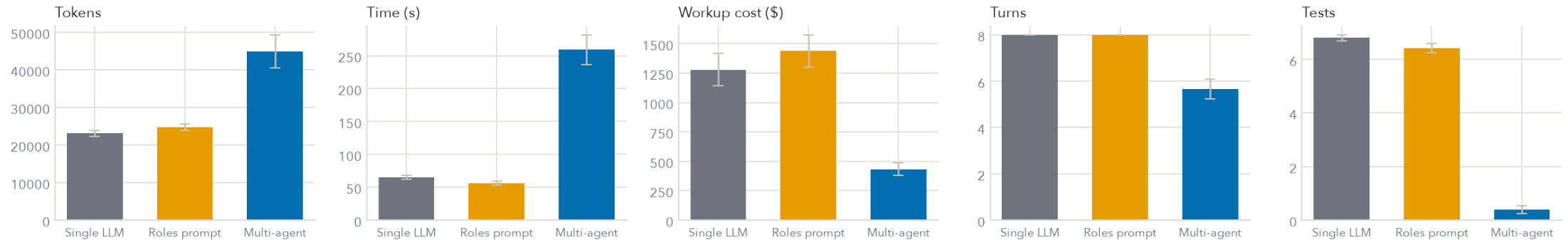
Design	Accuracy % (95% CI)	Judge Mean 1-5	Tokens Per Case	Time S / Case	Tests Per Case
Single LLM	23 ±8	1.80	23.1k	65	6.8
Roles prompt	23 ±8	1.90	24.7k	56	6.4
Multi-agent	13 ±7	1.47	44.9k	259	0.4

Holding the five roles fixed, more orchestration does not raise accuracy: the single call and the roles prompt are indistinguishable, and the multi-agent panel is lower while spending ≈1.9× the tokens and ≈4.3× the wall-clock time. Green dot marks the best value in each column.

THE COST OF COMPLEXITY

And the multi-agent panel costs far more

CPC – cost of complexity, per case (llama3.1-8b); error bars = 95% CI



The complex panel spends **~1.9× the tokens** and **~4.3× the wall-clock time** of a single call – for lower accuracy.

Robust across every checkpoint, $n = 10 \rightarrow 100$

Goals & timeline

- **Now:** full CPC design comparison done on an open 8B model – single $\approx$ roles, multi-agent weakest
- **Immediate:** reproduce $\sim$ 80% with frontier models; de-noise Judge/Gatekeeper; MIMIC-IV concordance
- **The contribution:** an adaptive delivery layer – when & how to surface each recommendation

PERIOD	MILESTONE	VENUE
Q1 · Sep '26	Benchmark & EHR environments	NeurIPS workshop
Q3 · Jan '27	Adaptive delivery policy	ICML (archival)
Q4 · May '27	Full system + clinical results	NeurIPS (archival)

Adaptive Alert Delivery